

Should I Go Out and Eat Friends with Friends Today?

Using Data to Start (or get out of) a Hangout

Project Overview and Motivation

Problem/Motivation

Going out in LA can be a pain

- One has to check weather and traffic (and others) to determine whether or not it is worth to go out

Most hangouts center around eating

- One has to check review sites (like Yelp) and scroll thru many reviews/ratings to determine whether or not the restaurant is worth going to

Proposed Solution

Create a dashboard that aids in making a decision to go out

- Acts as a central location to check weather, air quality, and traffic conditions

Create a model that aids in making a restaurant choice

- Use specific features of restaurants and create a classification model that determines if a restaurant is worth going to

Data Collection Approach and Progress

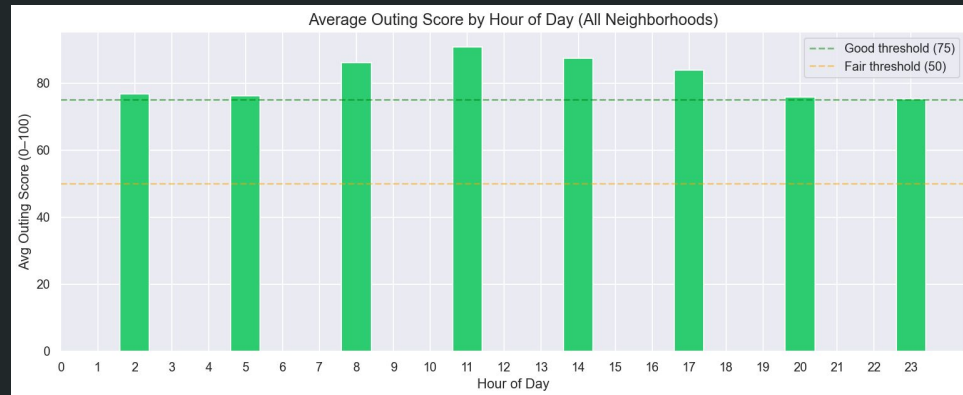
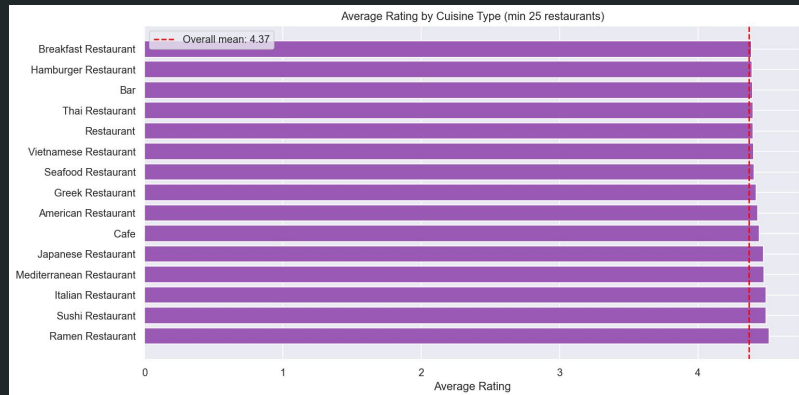
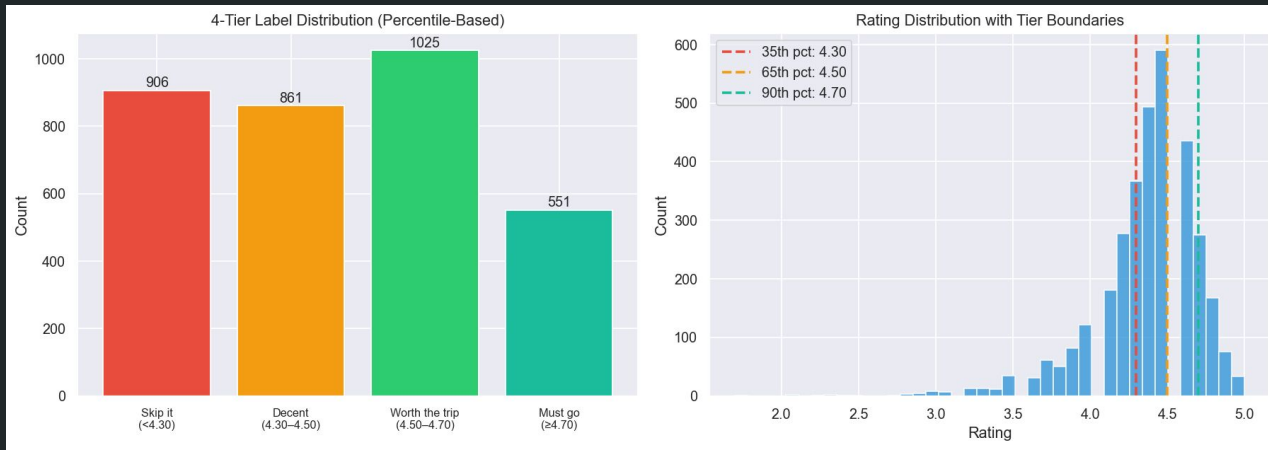
Conditions Data (Dashboard)

- Data will be collected with OpenWeatherMap and TomTom
 - OpenWeatherMap: current and forecast weather and air quality data via a live API
 - TomTom: live traffic data via a live API
- These APIs do not have historical data endpoints
 - Utilize Supabase and Github Actions to store hourly data thru a SQL server

Restaurant Data (Model)

- Data is collected with Google Places API and trained using multiple features

About the Data (So Far)



Modeling Approach

Target Variable: 4 tier restaurant quality classification based on percentile of ratings

- Skip it (0-35 percentile) / Decent (36-65) / Worth the trip (66-90) / Must go (90+)

Features: Review Count, Cuisine Type, Price Level, Reservable, Serves Alcohol, Provides Delivery/Dine In/Takeout, Popularity Flag, etc.

Modeling: XGBoost (Complexity and Performance) or Random Forest (Interpretability)

- Targeting a weighted F1 between 0.35 and 0.70, similar/balanced class distributions, and few distant misclassifications
 - "Skip it" -> "Must Go" vs "Skip It" -> "Decent"

What's Next?

Data Sources

Google Places API, OpenWeatherMap,
TomTom

Collection

Python Scripts + GitHub Actions Cron

Data Storage

CSV Storage + Supabase PostgreSQL

Flask API + Streamlit

Establish Connections and Build Dashboard to
Reach End User

Week 6/7: Establish
Database and Workflow
(Supabase + GitHub Actions)

Week 8/9: Build Flask API +
Streamlit App, Dockerize,
Deploy to Cloud Run

Week 10: Final Testing and
Official Deployment
